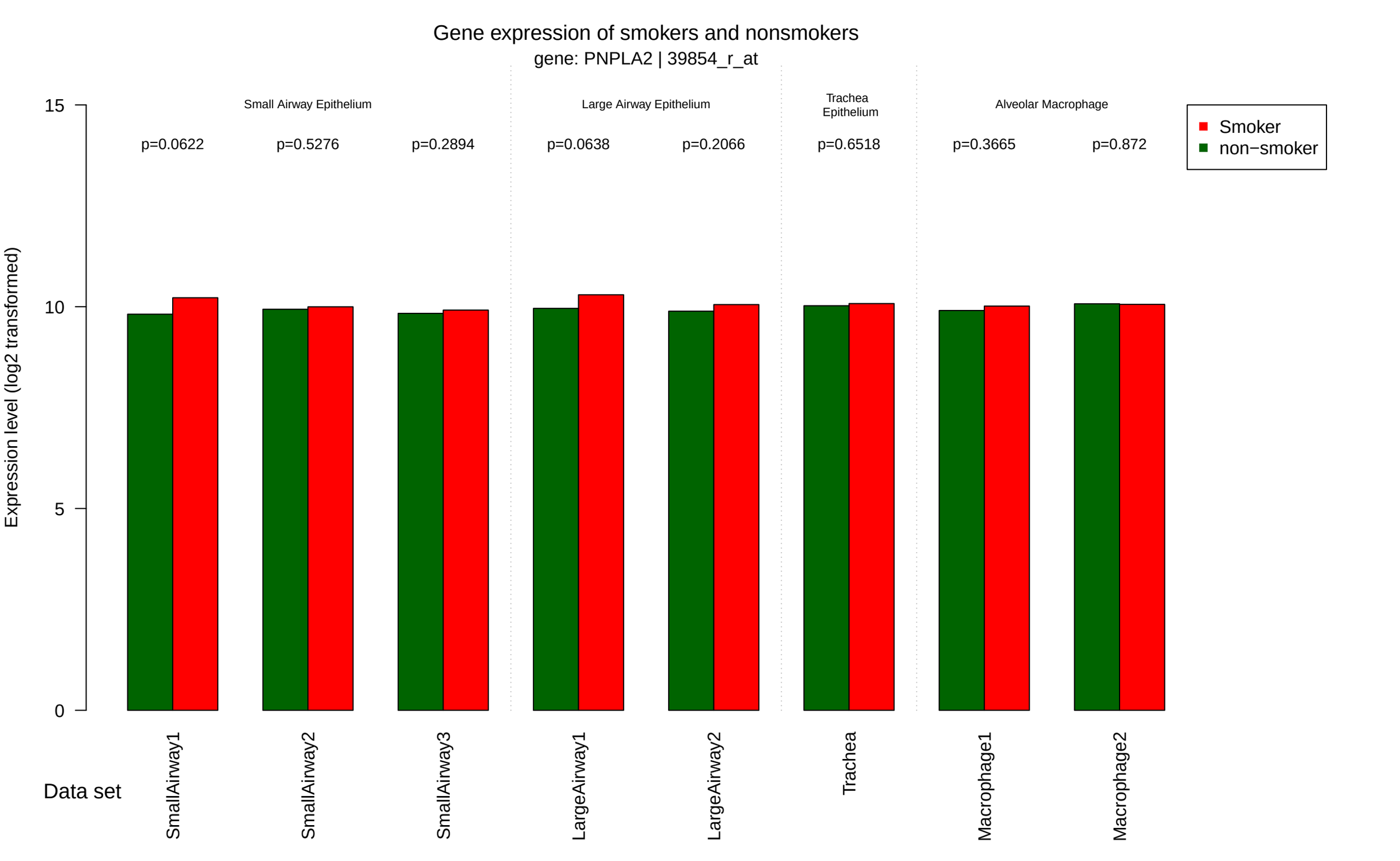
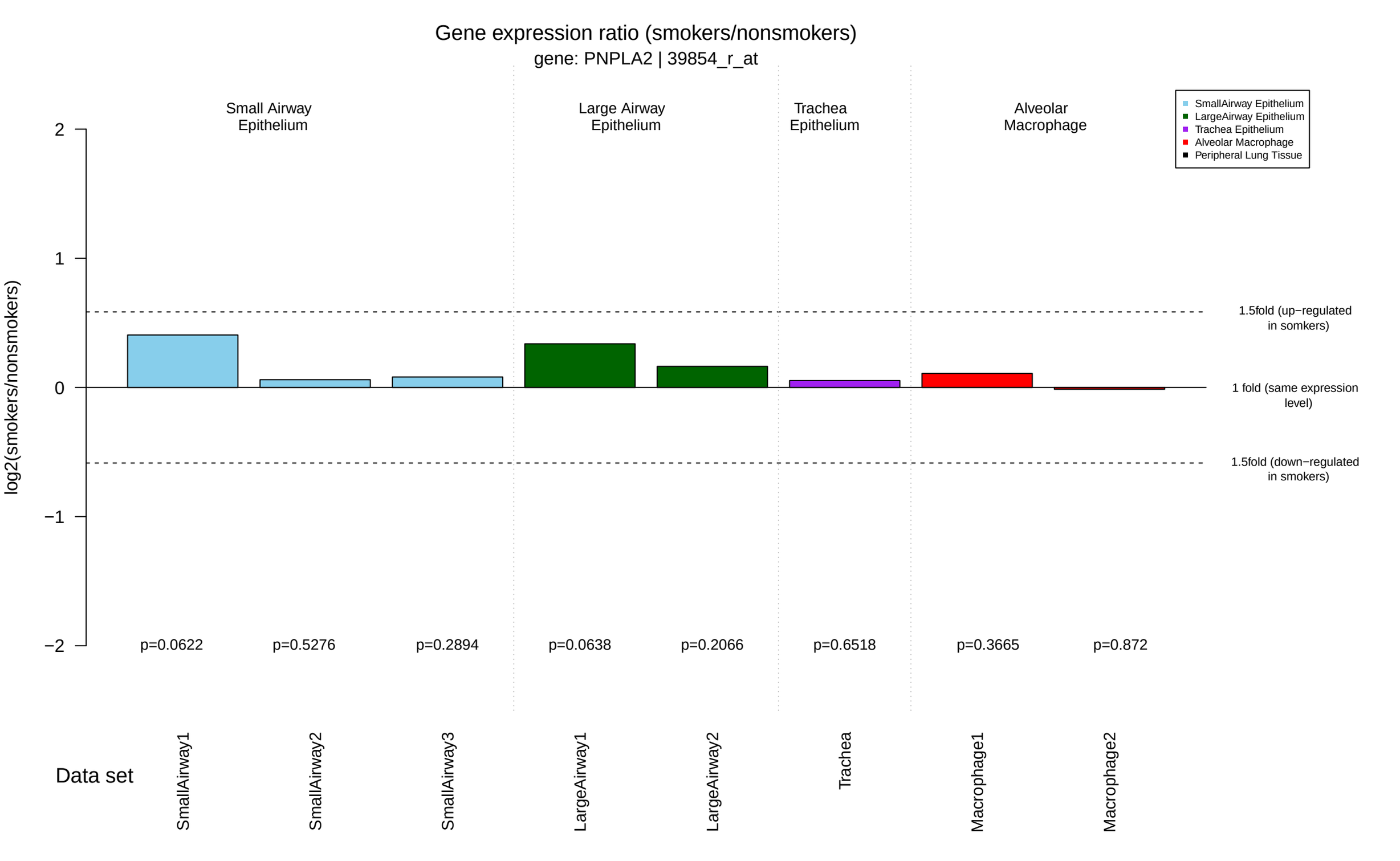
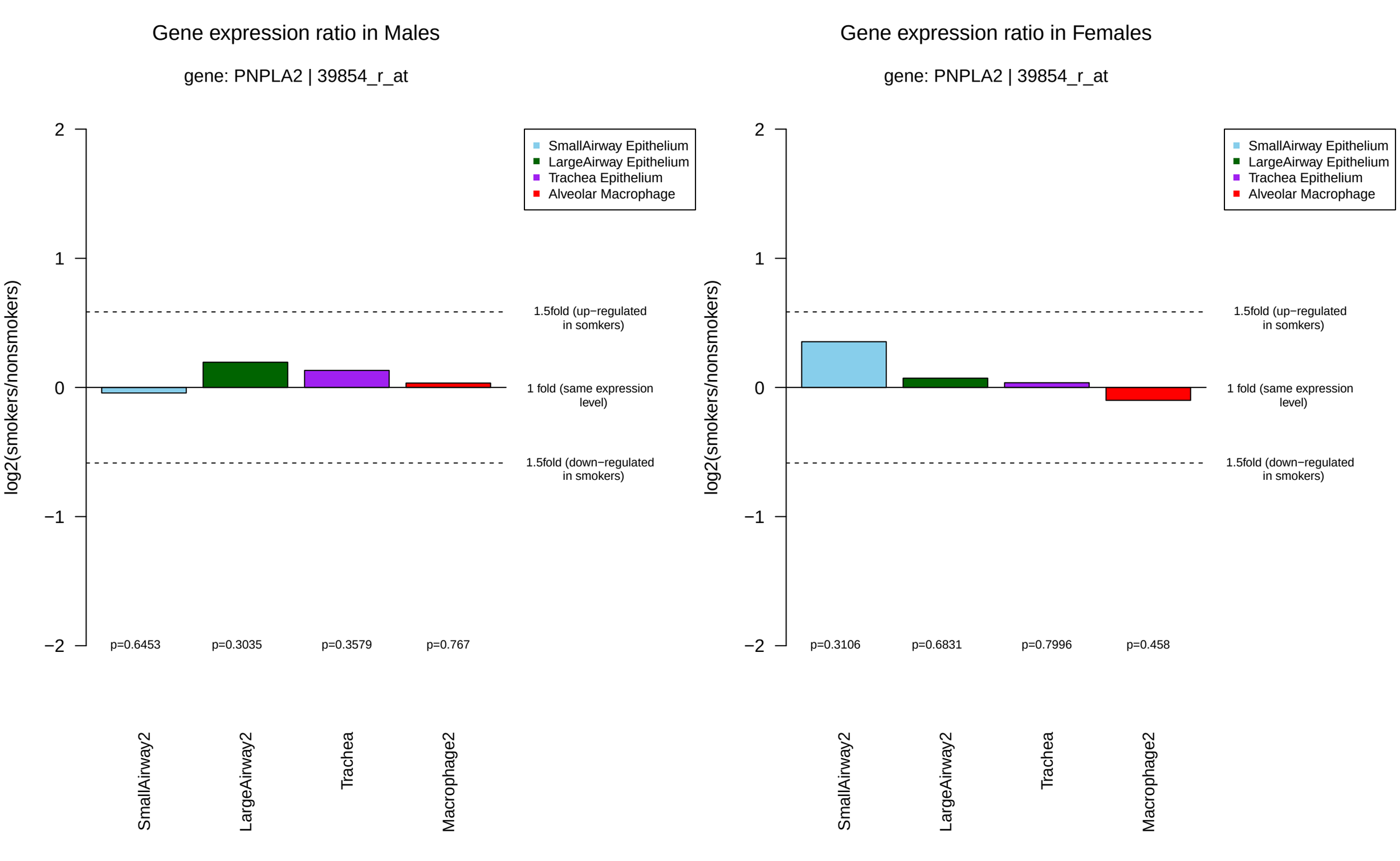
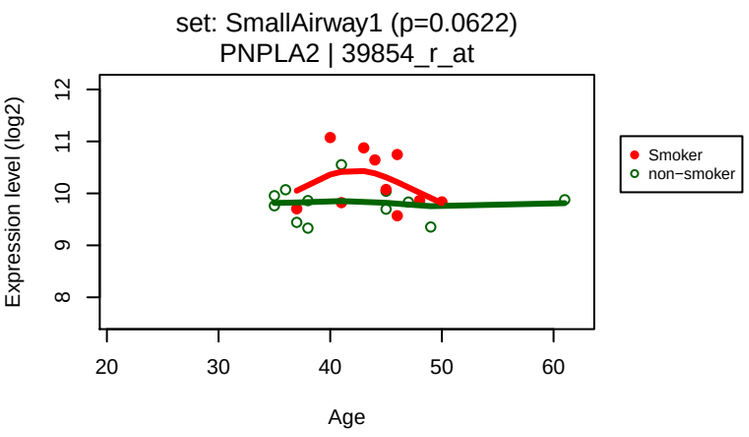
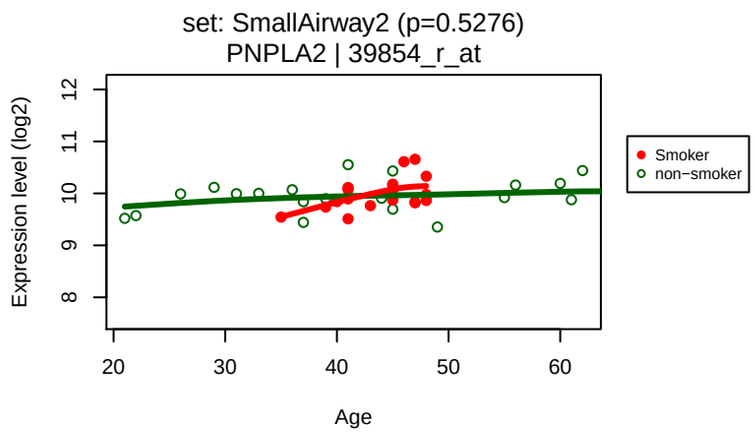
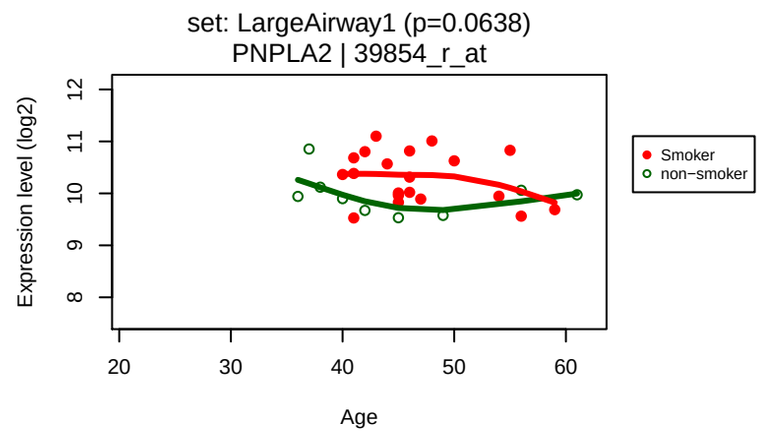

Gene expression of smokers and nonsmokers
gene: PNPLA2 | 39854_r_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0622
p=0.5276
p=0.2894
p=0.0638
p=0.2066
p=0.6518
p=0.3665
p=0.872
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

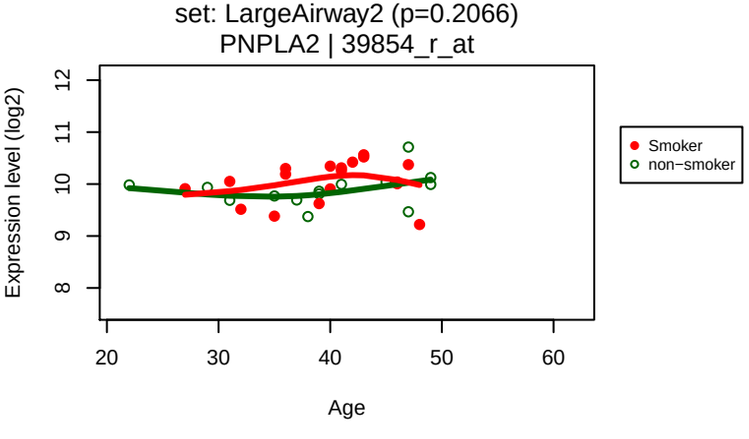

Gene expression ratio (smokers/nonsmokers)
gene: PNPLA2 | 39854_r_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0622
p=0.5276
p=0.2894
p=0.0638
p=0.2066
p=0.6518
p=0.3665
p=0.872
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

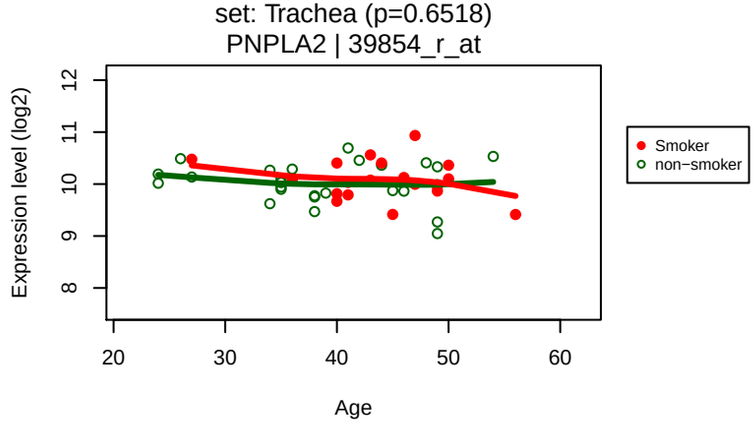

Gene expression ratio in Males
gene: PNPLA2 | 39854_r_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6453
p=0.3035
p=0.3579
p=0.767
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNPLA2 | 39854_r_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3106
p=0.6831
p=0.7996
p=0.458
SmallAirway2
LargeAirway2
Trachea
Macrophage2

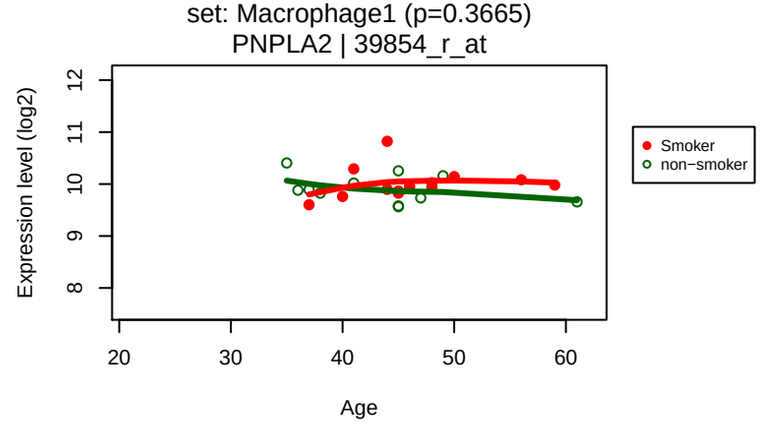

set: SmallAirway1 (p=0.0622)
PNPLA2 | 39854_r_at
Expression level (log2)
Age
Smoker
non−smoker

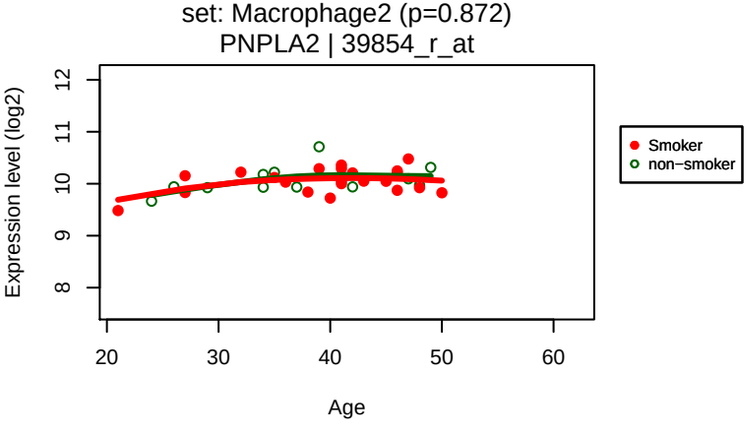

set: SmallAirway2 (p=0.5276)
PNPLA2 | 39854_r_at
Expression level (log2)
Age
Smoker
non−smoker

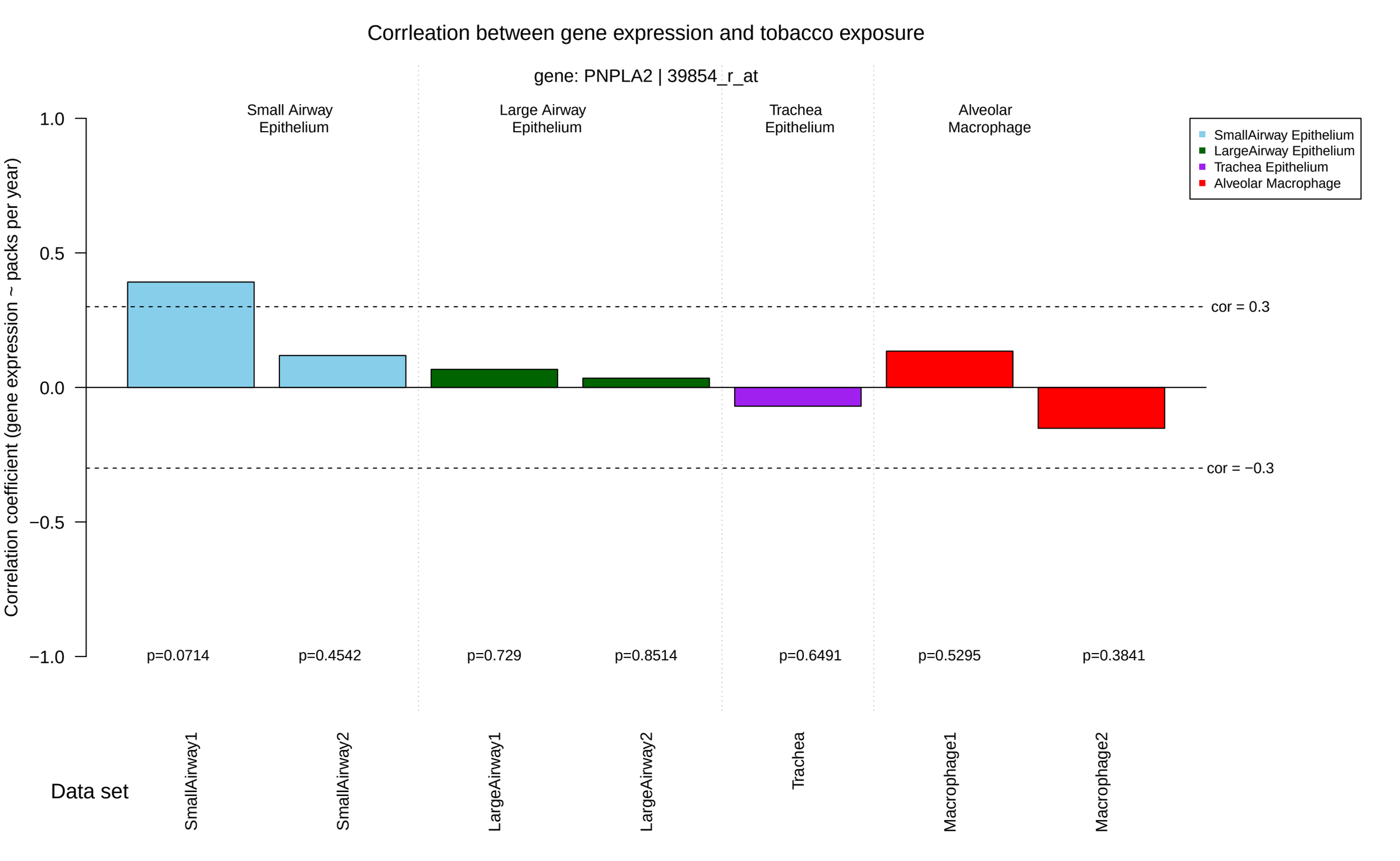

set: LargeAirway1 (p=0.0638)
PNPLA2 | 39854_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2066)
PNPLA2 | 39854_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6518)
PNPLA2 | 39854_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3665)
PNPLA2 | 39854_r_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.872)
PNPLA2 | 39854_r_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNPLA2 | 39854_r_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0714
p=0.4542
p=0.729
p=0.8514
p=0.6491
p=0.5295
p=0.3841
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2